

Problem 2

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Problem 1 State the form of the limit and then evaluate the limit.

$$(a) \lim_{x \rightarrow 6} \frac{4x^2 - 144}{x - 6}$$

Explanation. $\lim_{x \rightarrow 6} \frac{4x^2 - 144}{x - 6} = \lim_{x \rightarrow 6} \frac{4(x^2 - 36)}{x - 6}.$

Form: $\frac{0}{0}$

$$\begin{aligned} \lim_{x \rightarrow 6} \frac{4x^2 - 144}{x - 6} &= \lim_{x \rightarrow 6} \frac{4(x - 6)(x + 6)}{x - 6} \\ &= \lim_{x \rightarrow 6} 4(x + 6) \\ &= 4(12) = 48 \end{aligned}$$

$$(b) \lim_{x \rightarrow 6} \frac{x - 6}{\sqrt{2x - 8} - 2}$$

Explanation.

Form: $\frac{0}{0}$

$$\begin{aligned} \lim_{x \rightarrow 6} \frac{x - 6}{\sqrt{2x - 8} - 2} &= \lim_{x \rightarrow 6} \frac{x - 6}{\sqrt{2x - 8} - 2} \cdot \frac{\sqrt{2x - 8} + 2}{\sqrt{2x - 8} + 2} \\ &= \lim_{x \rightarrow 6} \frac{(x - 6)(\sqrt{2x - 8} + 2)}{2x - 8 - 4} \\ &= \lim_{x \rightarrow 6} \frac{(x - 6)(\sqrt{2x - 8} + 2)}{2(x - 6)} \\ &= \lim_{x \rightarrow 6} \frac{\sqrt{2x - 8} + 2}{2} \\ &= \frac{\sqrt{12 - 8} + 2}{2} \\ &= \frac{4}{2} = 2 \end{aligned}$$

$$(c) \lim_{x \rightarrow 2} \frac{(3x - 2)^2 - 16}{x - 2}$$

Explanation.

$$\begin{aligned}
 & \text{Form: } \frac{0}{0} \\
 \lim_{x \rightarrow 2} 2 \frac{(3x-2)^2 - 16}{x-2} &= \lim_{x \rightarrow 2} 2 \frac{((3x-2)-4)((3x-2)+4)}{x-2} \\
 &= \lim_{x \rightarrow 2} 2 \frac{(3x-6)(3x+2)}{x-2} \\
 &= \lim_{x \rightarrow 2} 2 \frac{3(x-2)(3x+2)}{x-2} \\
 &= \lim_{x \rightarrow 2} 23(3x+2) \\
 &= 3(6+2) = 24
 \end{aligned}$$

$$(d) \lim_{x \rightarrow 6} \frac{\frac{x}{x-2} - \frac{3}{2}}{x-6}$$

Explanation.

Form: $\frac{0}{0}$

$$\begin{aligned}
 \lim_{x \rightarrow 6} 6 \frac{\frac{x}{x-2} - \frac{3}{2}}{x-6} &= \lim_{x \rightarrow 6} 6 \frac{\frac{2x}{2(x-2)} - \frac{3(x-2)}{2(x-2)}}{x-6} \\
 &= \lim_{x \rightarrow 6} 6 \frac{\frac{2x-3x+6}{2(x-2)}}{x-6} \\
 &= \lim_{x \rightarrow 6} 6 \frac{\frac{-x+6}{2(x-2)}}{x-6} \\
 &= \lim_{x \rightarrow 6} 6 \frac{-(x-6)}{2(x-2)(x-6)} \\
 &= \lim_{x \rightarrow 6} 6 \frac{-1}{2(x-2)} \\
 &= \frac{-1}{2(6-2)} = \frac{-1}{2(4)} = \frac{-1}{8}
 \end{aligned}$$

$$(e) \lim_{x \rightarrow 1} \frac{\sqrt{5x-2} - \sqrt{3}}{x-1}$$

Explanation.

$$\begin{aligned}
 & \text{Form: } \frac{0}{0} \\
 \lim_{x \rightarrow 1} \frac{\sqrt{5x-2} - \sqrt{3}}{x-1} &= \lim_{x \rightarrow 1} \frac{\sqrt{5x-2} - \sqrt{3}}{x-1} \cdot \frac{\sqrt{5x-2} + \sqrt{3}}{\sqrt{5x-2} + \sqrt{3}} \\
 &= \lim_{x \rightarrow 1} \frac{(5x-2) - 3}{(x-1)(\sqrt{5x-2} + \sqrt{3})} \\
 &= \lim_{x \rightarrow 1} \frac{5(x-1)}{(x-1)(\sqrt{5x-2} + \sqrt{3})} \\
 &= \lim_{x \rightarrow 1} \frac{5}{\sqrt{5x-2} + \sqrt{3}} \\
 &= \frac{5}{\sqrt{5(1)-2} + \sqrt{3}} \\
 &= \frac{5}{2\sqrt{3}}
 \end{aligned}$$